

Highly Efficient Chip-Scale Package LED Based on Surface Patterning

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Abstract—Chip-scale package (CSP) light-emitting diodes (LEDs) are considered as the next generation of highly efficient package LED. However, they suffer from total internal reflection (LEE). In this letter, surface patterning, in terms of microstructures, as a well-controlled surface roughness method is designed and applied on CSP-LED to improve the LEE. Based on simulation results, pyramid and flat top pyramid structures have been designed and optimized. Four types of layouts, including tightly, interlaced, distant and chessboard layout, are also compared. A LEE enhancement of more than 20% is achieved. Furthermore, the four layouts with pyramid structures are fabricated using nano-imprint technology. The experimental results match the simulated results well with only a mismatch of 2.9%. The fabricated layout shows an enhancement of 20.31% when pyramid structures are fully filling the surface. Surface patterned CSP-LED, therefore, exhibits a great potential for high efficient LED.

Index Terms—Light emitting diodes, light extraction efficiency, nano-imprint, surface patterning.

I. INTRODUCTION

LIGHT emitting diode (LED) is regarded as one of the most important light sources in solid-state lighting owing to advantages in energy efficiency, long lifetime, vivid colors, high reliability, environmental protection, safety and multiple applications [1]–[5]. Package protects LED and ensures the service lifetime, meanwhile, package helps improving the electrical, thermal and optical performance [6]. As an emerging LED package technology, chip scale package LED (CSP-LED) is developing impressively in recent years. The kernel area and packaging area ratio of CSP-LED is around 1:1.2 to 1:1.7. It brings several advantages like compact size, low profile and low material cost. All these characteristics result in a

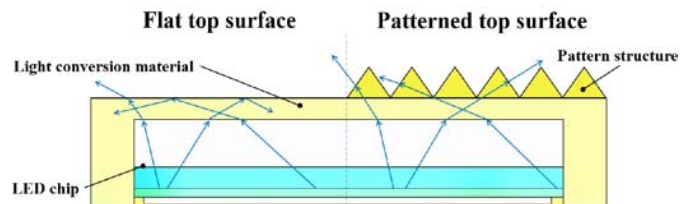


Fig. 1. Light trapped inside of light conversion material because of TIR and surface patterning helps the light extraction of LED.

high assembly density which satisfy the integration density and the output power required by the industrial applications. However, CSP-LED still faces the problem of low LEE (less than 30%) [7]. Limited LEE is mainly caused by the smooth interface between air and light conversion material. Besides, the large refraction index difference between the light conversion layer ($n \sim 1.6$) and air ($n = 1$) results in a small critical angle. Light incident into the interface beyond the critical angle will all be reflected back. A great amount of light is trapped inside of the silicone layer due to the total internal reflection (TIR) occurred here, as is shown in the left part of Fig. 1. This disadvantage stands in the way of improving the efficiency of CSP-LED.

In order to break this TIR and improve the LEE of LED chips, researchers have proposed several methods mainly focus on chip-shaped LED design and surface modification. The former includes the truncated inverted pyramid (TIP) LEDs [8] to maximize the top surface emitting. The latter including the random surface texturing applied to sapphire substrate [9], to the GaN surface [10], and to sidewalls [11] because of the easy access and acceptable enhancement. On the top of the emitting surface, microspheres arrays [12] optical diffractive elements such as micro-pyramids [8], ellipsoidal nanoparticles [13] and nanowire array [14] have also been used in a well-arranged way to meet an uniform distributed diffractive elements on the top surface. However, most of the approaches require multi-step processes including but not limited to film deposition, lithography, self-assembly, and dry etching or wet etching. These additional processing steps will surely increase the time and cost consuming.

Another cost-effective method to improve LEE of packaged LED is adding microstructures on top of package material to break the smooth interface between the air-package material. Compared with the flat top surface, the patterned top surface provides a decreased incident angle for light when reach at, and these light with incident angle beneath the critical angle

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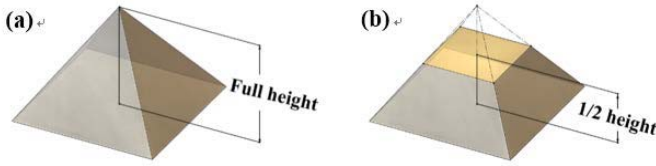


Fig. 2. (a) Pyramid and (b) flat top pyramid patterns.

will pass through, as is shown in the right part of Fig. 1. Simulation works [8], [15]–[17] based on Monte Carlo ray-tracing method, and experimental works have been carried out. Micro-fabrication methods including nano-imprint [18]–[20], laser patterning [19] and lithography [20] are utilized to achieve the designed structures. However, these mentioned approaches still require laser or vacuum which may increase the cost. An efficient and cost-effective design guideline for enhancement LEE is considerably desired for high performance CSP-LED package. Our work extends the scope of these studies by combining the Monte Carlo simulation and the nano-imprint technique by mold stamping to fabricate three dimensional structures to fully address the problem of the TIR induced low LEE of CSP-LED.

In this letter, we designed, simulated and optimized pyramid and flat top pyramid as well as tightly, interlaced, distant and chessboard layout. Then the best performance structure, pyramid structure, with all mentioned layouts was fabricated using wet etching and nano-imprint technology. The surface patterned CSP-LED (SPCSP-LED) all showed enhanced LEE, the best performance SPCSP-LED showed an optimized enhancement of 20.31%.

II. MODELING AND SIMULATION

A. Design of the Surface Pattern

To release the TIR light confinement effect, the designed patterns must reduce the number of light rays with incident angle larger than the critical angle at the interface between light conversion material and air. Several periodic patterns such as pyramid, flat top pyramid, spherical cap and the corresponding inverted structures were simulated to meet the requirement. Considered the fabrication cost and estimated performance, we selected pyramid and flat top pyramid array shown in Fig. 2.

Pyramid array was obtained by etching with the square pattern results in a pit with well-defined sidewalls (at angles of 54.7 degrees to the surface) [21]. From the base of the pyramid, we cut 1/2, 2/3, 3/4 top tip respectively to obtain 1/2, 1/3, 1/4 height flat top pyramids. Side length l of these patterns were designed to be $100\mu\text{m}$ or $200\mu\text{m}$, to cover the whole top surface of CSP-LED. As shown in Fig. 3, the package side length a is 1.2 mm which is the same size of actual CSP-LED.

On silicone surface of CSP-LED, four types of layouts have been designed. They are tightly, interlaced, distant and chessboard layout. All of them are showed in Fig. 4. Tightly layout arranges every pyramid adjoined with each other. Interlaced layout arranges the pyramids like the bricks in the wall, and b is the dislocation as shown in Fig. 4(b).

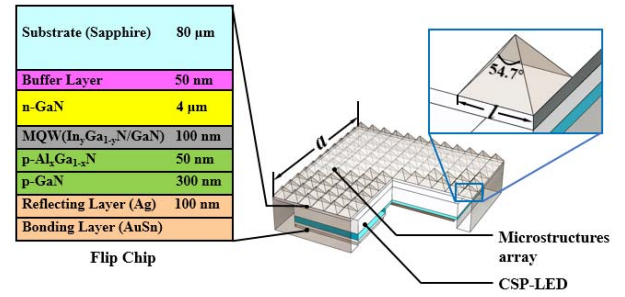


Fig. 3. The schematic of a SPCSP-LED.

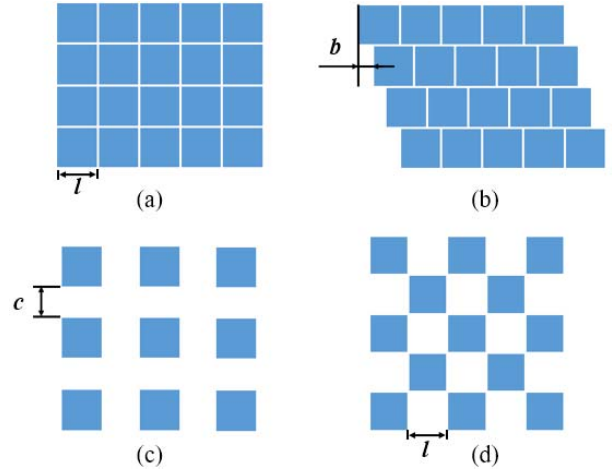


Fig. 4. Various layouts for both pyramids and flat-top pyramids including (a) tightly, (b) interlaced, (c) distant, (d) chessboard layout.

Distant layout indicates that pyramids are arranged with interval distance c . At last, chessboard layout means that the pyramids are separated by length l .

B. Establishment of the SPCSP-LED Model

The model of SPCSP-LED with flip chip LED is shown in Fig. 3. Flip chip LED schematic is included in the inset. Microstructures on the silicone or phosphor layer were integrated on the top of LED chip [22]. A model for SPCSP-LED was built including the substrate, reflecting layer, P-GaN, MQW layer, N-GaN, sapphire and the top layer of silicone or phosphor [14]. Layer thickness were included in Fig. 3. The optical parameters for each layer in SPCSP-LED were listed in the supporting document. The shape, size, separation and arrangement of the microstructures were discussed to find out how much the above parameters affect the LEE of SPCSP-LED.

C. Monte Carlo Ray-Tracing Simulation for Blue CSP-LED

The microstructures of pyramid and flat top pyramid were simulated with tightly, interlaced, distant and chessboard layout under the same simulation condition. Ten cases have been designed using the ray-tracing simulation, as is shown in Table I.

Photons were randomly generated at the MQW layer during the ray-tracing simulation, with their wavelength at 465 nm.

TABLE I
SIMULATION CASES FOR BLUE SPCSP-LED

Case	Layout	Side length (l) (μm)	Distance (b/c) (μm)
1	Tightly layout	100	-
2		200	-
3		100	10
4	Interlaced layout	100	20
5			50
6			10
7	Distant layout	100	20
8			50
9	Chessboard Layout	100	-
10		200	-

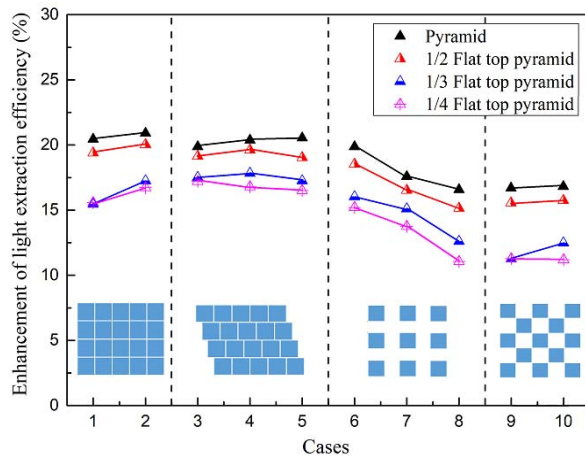


Fig. 5. Enhancement of LEE by adding surface patterned array of SPCSP-LED in the simulation.

The total number of rays is set as 4000000, and the threshold was set as 10^{-5} . The LEE was obtained from the simulation results. Compared with the blue CSP-LED with no surface structure, ten types blue SPCSP-LED have improved 10%~20% LEE that we can find out from Fig. 5. Among these four types of structures, the full height pyramid structure shows the best performance.

As is shown in Fig. 5, all flat top pyramid structures have poorer performance. The flat top could be considered as unpatterned surface, thus have no enhancement of the LEE. Considering this to all 10 cases with full height pyramid:

Cases (1–5) have similar enhancement because they have fully covered structures. Though the patterns are differently arranged, these cases have 100% covered pyramid on it. Case 2 performs better than case 1 because of its larger structure size. For cases (3–5), interlace distance of 10 μm and 20 μm would break the ordered arrangement, while case 5, with interlace distance of 50 μm , just returns to a well-ordered way. This kind of well-ordered arrangement helps light passthrough, rather than re-incident into light conversion material.

Incomplete-covered layouts, cases (5–10), have less structures compared with cases (1–5), thus have lower enhancement of LEE. The increasing gap in cases (6–8) results in less structure remains on the top surface of our CSP-LED. Two cases

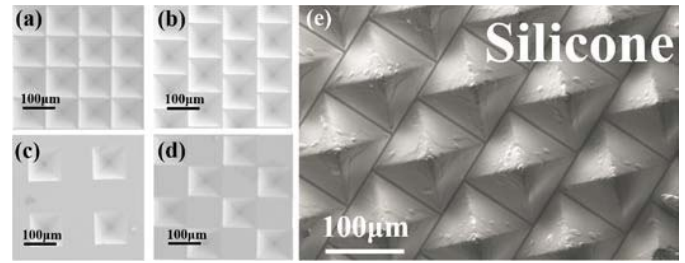


Fig. 6. SEM images of anti-pyramid molds: (a) tightly, (b) interlaced, (c) distant, (d) chessboard layout; and (e) silicone patterns.

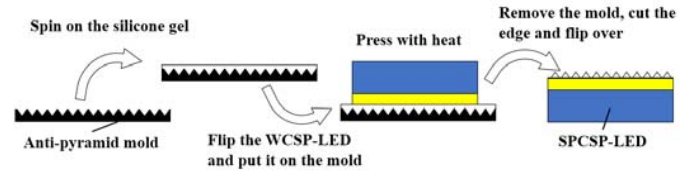


Fig. 7. Fabrication process of SPCSP-LED using nano-imprint technology.

for chessboard layout have similar enhancement because they have the same structure-cover-rate of 50%.

III. EXPERIMENT AND RESULTS

In order to further validate the feasibility of our surface patterning designs, we have fabricated blue SPCSP-LED with four kinds of surface patterning arrays according to the ray-tracing simulation results. Molds made of Si wafer were obtained by wet-etching and the pyramid structure is fabricated by nano-imprint technology.

Four types of anti-pyramid molds were fabricated using wet etching technology. Si wafers with different patterns on the top surfaces were immersed into etchant. By controlling the etching rate and time, we received these molds with acceptable performance, as is shown in Figs. 6(a–d). These molds are essential to the fabrication of SPCSP-LED.

The fabrication process flow of SPCSP-LED is shown in Fig. 7. First of all, silicone was spin-coated on our constructed anti-pyramid mold. The WCSP-LED was flipped upside down and set on the mold with silicone. Then nano-imprint instrument was adopted to apply pressure and heat. Silicone was cured at 150°C for 10 minutes. At last, the mold was removed carefully, the excess silicone edge was cut off and the SPCSP-LED was achieved. The silicone patterns had acceptable qualities shown in Fig. 6(e), the roughness on the patterned surfaces was caused by agent added between the mold and silicone to assist the separation.

Commercial samples of bare flip chip and CSP-LED were tested by ATA-500 auto-temperature LED opto-electronic analyzer from Everfine. LEE of the blue CSP-LED with no surface structure is shown in Fig. 8 and the CSP-LED with or without surface patterning are compared in this figure. It can be found that surface patterning has improved the LEE of CSP-LED about 10~20%.

All experimental results show less enhancement than the simulated ones, the fabrication error should take the responsibility. Except for the imperfect silicone surface shown above

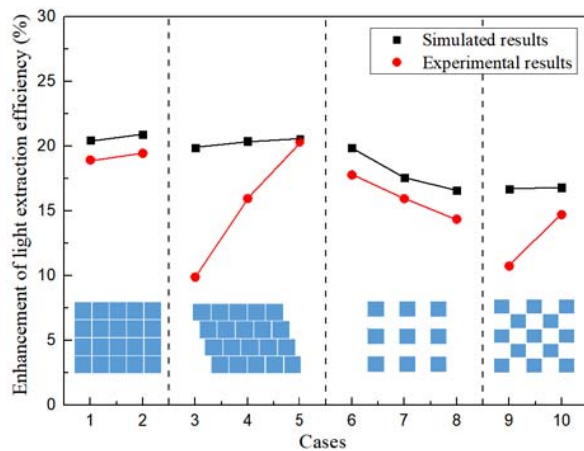


Fig. 8. Enhancement of LEE by adding pyramid array of various layouts on the top surface CSP-LED package.

in Fig.6(e), imperfect anti-pyramid mold is another reason for lower experimental performance. The lines contain no area in our designed pattern now turn into gaps with countable area, and this area with no structures on it makes the enhancement of LEE lower. Among all these cases, case 2 and case 5 have the most enhancement, which means tightly layout and interlaced layout are the best arrangement in these ten cases. There is 20.31% improvement of light extraction efficiency reached during case 5. There is only mismatch of 2.9% between experimental and simulated result for this case. In addition, the array whose pyramid side width is $200\ \mu\text{m}$ exhibits better result than the $100\ \mu\text{m}$ ones because they are easier to be achieved during the anti-pyramid mold fabrication process. For interlaced layout (cases 3–5), the interlace makes the fabrication of mold harder, especially for small interlace distance, this results in severely poor enhancements achieved by case 3 and 4. Meanwhile, for distant layout, when c is increased, the LEE reduced apparently. Therefore, the growth tendency is much sharp in experimental results. In general, experimental results show that the surface patterning process will enhance the LEE of CSP-LED and excellent design of patterns array will have an impressive promotion.

IV. CONCLUSION

Pyramid and flat top pyramid patterns, as well as layouts including tightly layout, interlaced layout, distant layout and chessboard layout have been designed and optimized using the Monte Carlo ray-tracing simulation method. Pyramid structures with all mentioned layouts were selected to continue further experiments using nano-imprint technology. Enhancement of LEE increased with increasing structure size, decreasing structure gap and decreasing fabrication error. Especially for SPCSP-LED had full height pyramid structures with side length of $100\ \mu\text{m}$, arranged as interlaced layout with dislocation of $50\ \mu\text{m}$, the LEE was enhanced by 20.31%, compared to the bare flat top CSP-LED. The best experimental results matched the simulated one well with only a mismatch of 2.9%. These results prove that surface patterning provides a predictable and effective solution to improve the

light extraction efficiency of CSP-LED during the packaging processing.

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